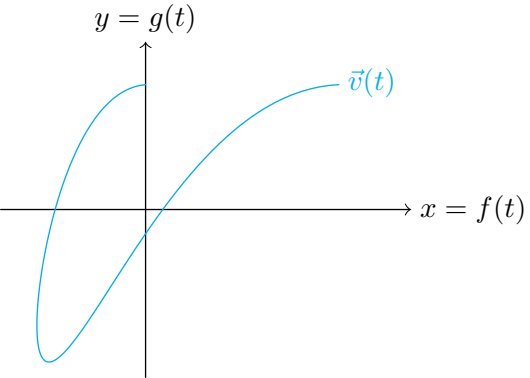


$$y = g(t)$$

$$\vec{v}(t)$$

$$x = f(t)$$


The image shows a 2D Cartesian coordinate system with a horizontal x-axis and a vertical y-axis. The x-axis is labeled $x = f(t)$ at its right end, and the y-axis is labeled $y = g(t)$ at its top end. A blue curve, labeled $\vec{v}(t)$ at its right end, is plotted. The curve starts in the third quadrant (negative x, negative y), crosses the x-axis at a positive value, crosses the y-axis at a positive value, and continues into the first quadrant (positive x, positive y). The curve is smooth and concave down in the first quadrant.